Igor Slazyk

ML Engineer | Data Scientist

Portfolio

islazykv.github.io

LinkedIn

linkedin.com/in/igor-slazyk

GitHub

github.com/islazykv

Personal Information

Nationality:	Polish
Date of Birth:	03/11/1992
Address:	Sandnes, Norway
Phone Number:	[redacted]
E-mail:	islazykv@gmail.com

Languages

English:	C1
Norwegian:	A2
Italian:	A2
Polish:	Native

Interests

Popular Science Literature
IT / Technology
Gaming
Cycling
Hiking

Summary

ML engineer with 8 years of experience in data science, including 4 years focused on machine learning. Passionate about turning data into valuable insights and impactful solutions. Driven to produce high-quality, scalable code and models. Collaborative team player, eager to contribute to projects at the intersection of AI and MLOps.

Education

Ph.D. in Particle Physics — 10/2017 – 07/2021

University of Ferrara — Ferrara, Italy
Institute of Nuclear Physics — Krakow, Poland

M.Sc. in Applied Physics — 02/2016 – 05/2017

Krakow University of Technology — Krakow, Poland

B.Sc. in Applied Physics — 09/2012 – 02/2016

Krakow University of Technology — Krakow, Poland

B.Sc. in Civil Engineering — 09/2011 – 02/2015

Krakow University of Technology — Krakow, Poland

Experience

Postdoctoral Research Fellow — 01/2022 – 05/2026

Høgskulen på Vestlandet — Bergen, Norway

- Development of ML models for data analysis in high-energy physics
- Design and implementation of scalable end-to-end ML pipelines
- Data preprocessing, feature engineering, model evaluation and optimization
- Data analysis and interpretation of results from ML-based physics searches
- Code maintenance, refactoring and automated testing
- Preparation of publications and presentation of ML results

Affiliated Researcher (ATLAS) — 01/2022 – 12/2025

CERN — Geneve, Switzerland

- Software development within the ATLAS experiment framework (Athena)
- Development of TauCP software for tau particle reconstruction and identification
- Code optimization, refactoring and integration into large-scale workflows
- Evaluation and interpretation of physics analysis results
- Collaboration with developers, analysts and physicists on software and data tools

Affiliated Researcher (LHCb) — 12/2017 – 03/2021

CERN — Geneve, Switzerland

- Software development for automated quality assurance of RICH detector modules
- Data analysis of quality assurance results for elementary detector cells
- Development of data processing scripts and tools for measurement workflows
- Hardware assembly, testing and quality control measurements
- Initial exploration of ML approaches for detector data analysis

Front-End Developer — 10/2016 – 09/2017

Freelance — Krakow, Poland

- Development of responsive websites using React.js
- Collaboration with back-end developers on feature integration and functionality

R&D Physics Intern — 04/2016 – 09/2016

Camlin Technologies — Zurich, Switzerland

- Data analysis of optical and quantum measurements of semiconductor materials
- Data collection and processing for photoluminescence spectroscopy
- Collaboration with software developers and physicists on analysis workflows

Selected Projects

Tau Supersymmetry Search

- repository
- notebook showcase
- publication

Boosted Decision Trees | Neural Networks

- Developed a supervised ML pipeline for multiclass classification with MLflow
- Trained and optimized classifiers with Optuna tuning
- Implemented ML-based regions construction and statistical hypothesis testing

Tau Anomaly Detection

- repository
- notebook showcase

Autoencoder | Variational Autoencoder

- Developed an unsupervised ML pipeline for anomaly detection with W&B
- Trained and optimized models with Ray Tune tuning
- Implemented reconstruction error anomaly scoring and latent space analysis

Tau Energy Scale

- internal project

Mixture Density Network | Neural Networks

- Developed a probabilistic regression model with uncertainty estimation
- Trained with Huber loss pretraining and NLL fine-tuning for density estimation
- Evaluated model performance with energy response and resolution histograms

Sphalerons and Black Holes

- publication

Convolutional Neural Network | Boosted Decision Trees

- Optimized high-capacity models to classify rare high-multiplicity LHC events
- Developed a pipeline mapping detector signals into three-layer event images
- Compared high-level features with raw pixel data to maximize signal sensitivity

Di-Tau Reconstruction

- internal project
- note

Software Development | Data Science

- Engineered a C++ analysis framework for boosted di-tau reconstruction studies
- Benchmarked fixed & variable radius subjets to optimize reconstruction efficiency
- Evaluated diverse seed jet collections to establish baseline configurations

Elementary Cells

- repository
- publication

Software Development | Data Science

- Developed an automated C++/Python/LabVIEW software for photomultiplier QA
- Analyzed array performance via noise, dark count and threshold measurements
- Explored ML-based particle identification for the LHCb RICH detector

Technical Skills

Programming Languages & Tools:

- Python, C++, SQL, Bash
- Git, Jupyter, pytest, uv, conda

Data Processing & Visualization:

- NumPy, pandas, SciPy, ROOT, uproot, Awkward Array, pyhf
- Matplotlib, seaborn, Plotly

Machine Learning & Deep Learning:

- scikit-learn, XGBoost
- PyTorch, PyTorch Lightning, TensorFlow, Keras

Tuning & Explainability:

- Optuna, Ray Tune, SHAP

MLOps & Deployment:

- MLflow, W&B, DVC, Hydra
- FastAPI, Docker, GitHub Actions

Web Development:

- HTML, CSS, JavaScript, React

Currently Learning:

- Kubernetes
- Azure ML
- Azure DevOps Pipelines
- Databricks